

Juan David Muñoz Bolaños

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jmunozbolanos@gmail.com

21. April 2026

SmarAct Metrology GmbH & Co. KG

z. Hd. Melissa Skrivanek

Oldenburg

Deutschland

Bewerbung als Applikationsingenieur (m/w/d) – Laserinterferometrie und Vibrometrie

Sehr geehrte Frau Skrivanek,

mit großem Interesse bewerbe ich mich auf die ausgeschriebene Stelle als Applikationsingenieur im Bereich Laserinterferometrie und Vibrometrie bei SmarAct Metrology. Als Physikingenieur mit Promotion in adaptiver Optik und Mikroskopie bringe ich genau die Kombination aus fundiertem Optikwissen, praxisnaher Prototypenentwicklung und starker Softwarekompetenz mit, die diese Position erfordert.

Im Kern meiner Doktorarbeit an der Medizinischen Universität Innsbruck steht der Aufbau eines interferometrischen Wellenfrontsensors für die adaptive Optik-Mikroskopie. Dabei habe ich das gesamte System eigenständig konzipiert und realisiert: vom optischen Aufbau eines Michelson-basierten Interferometers über die Implementierung iterativer Phasenrekonstruktionsalgorithmen bis hin zur vollständigen Steuerung in Python, C++ und LabVIEW. Dieses System ist täglich im Einsatz und bildet die Grundlage zweier Publikationen in ACS Photonics und Biomedical Optics Express. Darüber hinaus habe ich ein komplettes Multiphotonen-Mikroskop von Grund auf aufgebaut – einschließlich Laserintegration, Galvo-Scanner-Steuerung und PMT-Signalerfassung – und systematische Toleranzanalysen in ZEMAX sowie Wellenfeldmodellierungen in Python durchgeführt.

Bereits vor meiner Promotion habe ich am Leibniz-Institut für Photonische Technologien (IPHT) in Jena einen dispersiven 1064 nm Faser-Raman-Spektrometer-Prototyp entwickelt, kalibriert und publiziert. Diese Erfahrung im vollständigen Prototypenzyklus – von der optischen Auslegung über die Validierung bis zur wissenschaftlichen Dokumentation – deckt sich direkt mit dem bei SmarAct geforderten Übergang von der Entwicklungsphase zur Serienproduktion. Ergänzend bringe ich industrielle Praxiserfahrung als Machine Vision Engineer mit, wo ich unter anderem optische Inspektionssysteme konzipiert und in die Produktion überführt habe.

Mein frühestmöglicher Eintrittstermin ist Mai 2026. Als Inhaber einer österreichischen Aufenthaltsgenehmigung bin ich uneingeschränkt arbeitsberechtigt. Ich freue mich darauf, meine Erfahrungen in Laserinterferometrie und optischer Messtechnik bei SmarAct einzubringen und stehe für ein persönliches Gespräch gerne zur Verfügung.

Mit freundlichen Grüßen,

Juan David Muñoz Bolaños

Doktorand, Medizinische Universität Innsbruck

Juan David Muñoz Bolaños

Application Engineer – Laser Interferometry & Optical

Metrology

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145

jmunozbolanos@gmail.com | LinkedIn Profile



Summary

Physicist and optical engineer with deep expertise in **laser interferometry, wavefront sensing, and precision optical metrology**. Proven track record in end-to-end prototype development – from optical concept and simulation (ZEMAX, Python) through hardware-software integration (Python, C++, LabVIEW) to validated, publication-grade systems. Strong analytical skills and experience transitioning R&D concepts into robust, reproducible measurement setups. Author of three peer-reviewed publications in applied optics and photonics.

Professional Experience

Jan'22 | Present **Med. Univ. Innsbruck**

Innsbruck, Austria

Optical Systems Engineer | PhD Researcher

Project 1 | Precise Aberration Correction for Confocal Microscopy:

Designed and implemented a closed loop optical aberration correction system. Applied advanced optical physics and Python based control to achieve diffraction limited optical perfection in complex scattering media (ACS Photonics 2025).

Project 2 | Multi-Device System Integration for Adaptive Optics: Led the optical design and integration of an aberration corrected two photon microscope. Orchestrated a complex optoelectronic stack including FPGA, LabVIEW, high speed galvos, lasers, Arduino microcontrollers, and electronic synchronization for high performance imaging.

Project 3 | Fast phase light modulator: Developed an automated interferometric metrology platform using one of the fastest holographic modulators. Integration via a C++, Python, and LabVIEW stack to recover image fidelity.

Jun'20 | Dec'21 **Leibniz Institute of Photonic Technology**

Jena, Germany

Assistant Researcher | Photonics & Spectroscopy

Dispersive Spectrometer Design: Led optical layout and microscope imaging integration for a 1064 nm fiber probe Raman spectrometer. Handled basic ZEMAX tolerancing, detector alignment, and wavelength calibration (Applied Sciences 2024).

Hyperspectral ML Classification: Developed Python pipelines for supervised machine learning on 2D Raman tissue images, achieving high accuracy cancer vs healthy cell separation.

Mar'18 | Mar'19 **INTECOL S.A.S.**

Medellín, Colombia

Junior Machine Vision System Engineer

Industrial Vision Pipelines: Deployed C++ inspection systems on high speed production lines. Engineered custom bright field, dark field, and ring light schemes resolving verification failures.

Hardware Setup: Selected industrial cameras, lenses, and sensors.

Implemented and validated optical inspection setups on live production lines.

Technical Competencies

Laser Interferometry & Metrology: Wavefront sensing, phase retrieval, Michelson interferometry, adaptive optics, precision alignment of laser-based measurement systems.

Prototyping & Optical Design: End-to-end system build from concept to validated prototype. ZEMAX/OpticStudio modelling, tolerance analysis, opto-mechanical integration, component qualification (Thorlabs, Edmund Optics, Hamamatsu).

Data Acquisition & Simulation: Python, C++, LabVIEW/FPGA, MATLAB for real-time data capture, signal processing, physical modelling, and instrument control. Experience with JAX (GPU computing).

Technical Documentation: Three peer-reviewed publications (ACS Photonics, Biomedical Optics Express, Applied Sciences). Experienced in writing application notes, technical specifications, and whitepaper-style documents.

Industrial Optics: Halcon & Cognex VisionPro pipelines, custom illumination design, camera integration, Gage R&R, system commissioning on production lines.

Education

Jan'22 Jun'26	Medical University of Innsbruck <i>Doctor of Philosophy in Adaptive Optics & Precision Metrology</i>	Innsbruck, Austria
2019 2021	Friedrich Schiller Univ. Jena <i>Master of Science in Photonics</i>	Jena, Germany
2012 2018	University of Cauca <i>Bachelor of Science in Physics Engineering</i>	Popayán, Colombia

Publications & Awards

Awards: Selected for ASML, ZEISS SMT, and SPIE (San Francisco) training programs. COLFUTURO 2016-2018.

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills & Hobbies

Project Ownership: Developed systems from optical design to prototypes and independently built the software for data analysis such as SpectraMap package pypi or custom phase retrieval algorithms.**Interdisciplinary Teamwork:** Collaborated with research and hardware teams across INTECOL, IPHT Jena, and Med. Univ. Innsbruck.**Systematic Problem Solving:** Diagnosed and resolved systematic production defects at INTECOL.**Hobbies:** Mountain biking, Bachata, Harmonica, Vision systems (cameras).

Languages

English (Fluent) · German (Intermediate B2) · Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
Medical University of Innsbruck
monika.ritsch-marte@i-med.ac.at

Prof. Dr. Alexander Jesacher
Medical University of Innsbruck
alexander.jesacher@i-med.ac.at

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

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Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

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Jena, 13 November 2020


Prof. Dr. Jürgen Popp
Chairman
of Leibniz-IPHT


F. Sondermann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

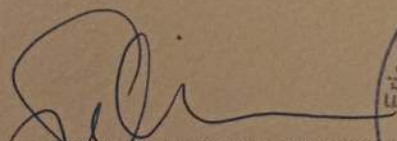
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy



ASML

PhD Master Class

Dear participant of the ASML PhD Master Class | Spring edition 2026,

We're looking forward to meeting you on March 19, 20 & 21.

Here's an information package containing details about the masterclass program. Please read this carefully so you know what to expect and how to prepare for it.

To get the most out of this masterclass for you, we want your input on what you would like to learn. At the end of this briefing, there is a link where we ask you to share your preferences with us.

Do not hesitate to contact us if you still have questions after this briefing.

Laura de Swart
Program Manager, ASML
+31 6 1562 7555

You will be joining us live at ASML HQ in Veldhoven on Friday and Saturday. Make sure you bring your **valid ID** to enter the ASML buildings. Without it, you will **not** be allowed inside. On Thursday we will meet at the Philips Stadium in the center of Eindhoven for an informal start to the event. The dress code for the entire masterclass is informal.

Program

Below is the program for the three days again. On the next pages, you will find more details on the different parts of the program.

PhD Master Class

Thursday, March 19

19:30 – 22:30 Informal activity @ Philips Stadium

Friday, March 20

08:30 – 09:00	Welcome
09:00 – 10:30	Introduction into ASML Technology
10:30 – 10:50	Break
10:50 – 12:00	Pitches from former participants
12:00 – 13:00	Lunch break
13:00 – 18:00	Working on cases
18:00 – 21:45	Dinner & presentations & feedback on case results
21:45 – 22:00	Closing day one

Saturday, March 21

08:30 – 09:00	Welcome back
09:00 – 12:50	Carousel • Matching conversations • HR corner • Q&A • ASML tours
12:50 – 13:50	Lunch break
13:50 – 14:00	Wrap up and closing

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
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Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532

Passaporto № / Passport No.

MUNOZ BOLANOS

COLOMBIAN

Sexo / Sex
Lugar de nacimiento / Place of birth

11 DIC/DEC 2018

Fecha de vencimiento / Date of expiry

10 DIC/DEC 2028

ANTIOQUILLA

Forma del budar / *Perispor* a *segru*

Dear Family

[illegible]

Stadtmagistrat Innsbruck
MA II - Bezirks- und Gemeindeverwaltung
Standesamt und Personenstandsangelegenheiten
Aufenthaltsangelegenheiten
post.aufenthaltsangelegenheiten@innsbruck.gv.at

Zahl: Maglbk/58566/AUFG-NAG/3/1 – 00053337-04

Antragsprotokoll

Herr: **Muñoz Bolaños Juan David**
wohnhaft in: **6020 Pradl, Reichenauer Straße 42/Top 5**
geboren am: **04.06.1994**
Reisedokument-Nr.: **AV410884**
Nationalität: **Kolumbien**
hat am: **13.03.2026**

beim Stadtmagistrat Innsbruck, Referat Aufenthaltsangelegenheiten, einen Zweckänderungsantrag auf Rot-Weiß-Rot - Karte plus eingebracht.

Ein rechtzeitig eingebrachter Verlängerungsauftrag räumt während der gesamten Dauer des Verlängerungsverfahrens dieselbe Rechtsposition ein, die nach Erhalt des letzten Aufenthaltstitels innehabt wurde. Dies bedeutet auch, dass bei befristet erteilten Aufenthaltstiteln mit unbeschränkten Zugang zum Arbeitsmarkt („Rot-Weiß-Rot Karte plus“ und „Familienangehöriger“) weiterhin während der gesamten Dauer des Verlängerungsverfahrens ein Zugang zum Arbeitsmarkt besteht. Bei darüber hinaus gehenden Fragen einer allfälligen weiterhin bestehenden Beschäftigung wenden Sie sich bitte an das Arbeitsmarktservice (AMS).

Nachzureichende Unterlagen: ***

Die Erledigung des vollständig eingebrachten Antrages wird voraussichtlich innerhalb von Wochen erfolgen.

Persönliche Abholung (erforderlich !) jedenfalls erst ab: 17.05.2026, Zimmer: .Nummer ziehen

Zur Abholung (Abholzeiten: Montag bis Freitag, 08.00 bis 11.30 Uhr) sind mitzubringen:

- **der Reisepass**
- **Aufenthaltstitel in Kartenform**

Für den Bürgermeister:

(Huter)

ausstellende Behörde:

Stadtmagistrat Innsbruck
Abteilung II
Aufenthaltsangelegenheiten

übernommen am: 13.03.2026

Unterschrift:

Bei überprüfenden Amtshandlungen drei Monate nach Ausstellung dieser Bestätigung wird um Rückfrage bei der zuständigen Aufenthaltsbehörde ersucht. (Tel.: 0512/5360-1030/1032/1034/1036)